

The Silent Tax: Typo-Robustness Analysis (GSM8K)

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All tables are *answered-only*: traces that hit the token cap before writing a final answer are excluded from the metrics. Configs are named **typo{rate}_real{ratio}**: *rate* = % of words corrupted, *real* = % of typos that form real words. Baseline is the clean input.

Accuracy and Flips

Table 1: Accuracy per config: answered-only and strict (truncated=wrong), with 95% bootstrap CI on acc_strict.

config	n	acc_answered	acc_strict	ci_lo	ci_hi
clean	500	0.9158	0.914	0.888	0.936
typo25_real10	500	0.8377	0.836	0.802	0.868
typo25_real40	500	0.829	0.824	0.79	0.856
typo25_real70	500	0.8016	0.8	0.764	0.836
typo50_real10	500	0.7635	0.762	0.726	0.798
typo50_real40	500	0.7154	0.714	0.674	0.752
typo50_real70	500	0.6794	0.678	0.634	0.716
typo75_real10	500	0.6687	0.658	0.616	0.7
typo75_real40	500	0.638	0.624	0.58	0.664
typo75_real70	500	0.5691	0.56	0.516	0.604

Table 2: Answered-only flips vs clean, paired by question over n_both (answered in both conditions). $r2w$ / $w2r$ = questions correct on clean but wrong on this config / wrong on clean but correct on this config; $r2w_ratio = r2w / n_both$, $w2r_ratio = w2r / n_both$; $net_flip_ratio = r2w_ratio - w2r_ratio = (r2w - w2r) / n_both$, the net rate at which this config worsens answers relative to clean.

config	n_both	r2w	w2r	r2w_ratio	w2r_ratio	net_flip_ratio	mcnemar_b	mcnemar_c	mcnemar_stat	mcnemar_p
typo25_real10	498	50	12	0.1004	0.0241	0.0763	50	12	22.081	2.61e-06
typo25_real40	496	55	12	0.1109	0.0242	0.0867	55	12	26.328	2.9e-07
typo25_real70	498	64	8	0.1285	0.0161	0.1124	64	8	42.014	0.0
typo50_real10	498	88	12	0.1767	0.0241	0.1526	88	12	56.25	0.0
typo50_real40	498	111	12	0.2229	0.0241	0.1988	111	12	78.081	0.0
typo50_real70	498	126	9	0.253	0.0181	0.2349	126	9	99.674	0.0
typo75_real10	491	130	10	0.2648	0.0204	0.2444	130	10	101.15	0.0
typo75_real40	488	151	17	0.3094	0.0348	0.2746	151	17	105.292	0.0
typo75_real70	491	183	12	0.3727	0.0244	0.3483	183	12	148.205	0.0

McNemar test (used in the flips table above).

- **What it is:** a paired significance test for before/after binary outcomes. It looks only at the questions that *changed* (the discordant pairs: b = right→wrong, c = wrong→right) and ignores those that stayed the same.

- **Null hypothesis:** typos break correct answers as often as they fix wrong ones ($b = c$, i.e. no net effect).
- **Statistic:** $(|b - c| - 1)^2 / (b + c)$ (column `mcnemar_stat`); `mcnemar_p` is its chi-square p -value with 1 degree of freedom.
- **When we reject it:** when `mcnemar_p` < 0.05 — the right→wrong vs wrong→right imbalance is too large to be chance.

Reasoning Length

Table 3: Absolute generated tokens per config (answered-only).

config	n_answered	mean_tok	median_tok	p90_tok	max_tok
clean	499	1325.8	1059	2498	10814
typo25_real10	499	1284.4	765	2290	13012
typo25_real40	497	1244.4	816	2517	13670
typo25_real70	499	1290.1	785	2436	18797
typo50_real10	499	1316.2	716	2815	12365
typo50_real40	499	1341.1	729	2657	14424
typo50_real70	499	1338.8	734	3185	11591
typo75_real10	492	1612.0	789.5	4396	12037
typo75_real40	489	1519.9	736	3957	9788
typo75_real70	492	1518.1	732.0	3682	13459

Table 4: Mean generated tokens for correct vs wrong answers. `wrong_over_correct` = `tok_wrong` / `tok_correct` (mean tokens of wrong answers divided by mean tokens of correct answers).

config	tok_correct	tok_wrong	wrong_over_correct
clean	1207.5	2613.2	2.1642
typo25_real10	1100.1	2235.6	2.0322
typo25_real40	1086.7	2008.6	1.8484
typo25_real70	1073.0	2167.2	2.0198
typo50_real10	1110.3	1981.3	1.7845
typo50_real40	1022.2	2142.6	2.096
typo50_real70	1065.3	1918.2	1.8006
typo75_real10	1224.1	2395.0	1.9565
typo75_real40	1043.8	2359.0	2.26
typo75_real70	1132.7	2027.1	1.7895

Self-doubt

The self-doubt markers are curated words grouped into two *banks*, each matched as a case-insensitive whole-word pattern in the reasoning text. Density is matches per 1000 reasoning words.

- **second_guess** — the proposal’s canonical self-correction words plus direct reversals: *wait, hmm, actually, reconsider, but wait, hold on, on second thought, let me recheck, double-check*.
- **uncertainty** — hedging / “not certain” words: *maybe, perhaps, possibly, might be, could be, not sure, I guess, presumably, or maybe / alternatively, unclear / confused*.

Example reasoning: “*Wait, maybe the total is 12... hmm, actually let me recheck*” contributes *wait*, *hmm*, *actually* (second_guess) and *maybe* (uncertainty). A separate *typo-noticing* bank (*typo*, *misspelled*, *doesn't make sense*, ...) is treated as repair behaviour, not self-doubt.

Table 5: Doubt-marker density per config: second_guess and uncertainty, per 1k words and per trace.

config	n.answered	second_guess_per_1k	second_guess_per_trace	uncertainty_per_1k	uncertainty_per_trace	total_per_1k	total_per_trace
clean	499	12.641	8.886	6.985	4.91	19.626	13.796
typo25_real10	499	12.603	8.174	11.327	7.347	23.93	15.521
typo25_real40	497	11.875	7.612	10.033	6.431	21.908	14.042
typo25_real70	499	12.6	8.283	11.524	7.575	24.124	15.858
typo50_real10	499	11.926	7.898	12.092	8.008	24.018	15.906
typo50_real40	499	11.576	7.968	14.744	10.148	26.321	18.116
typo50_real70	499	12.574	8.685	14.158	9.78	26.732	18.465
typo75_real10	492	12.991	11.1	16.314	13.939	29.305	25.039
typo75_real40	489	12.226	9.867	18.267	14.742	30.493	24.609
typo75_real70	492	11.577	9.295	20.985	16.848	32.562	26.142

Table 6: Per-marker density (markers per 1000 reasoning words): clean vs pooled-typo. discrimination = typo minus clean (larger = more typo-responsive). All bank markers, ranked by discrimination.

marker	category	clean_per_1k	typo_per_1k	discrimination
maybe	uncertainty	3.279	7.434	4.155
or maybe/alt	uncertainty	1.85	3.322	1.472
wait	second_guess	6.127	7.542	1.415
perhaps	uncertainty	0.807	1.99	1.183
unclear/confused	uncertainty	0.676	1.064	0.388
might be	uncertainty	0.08	0.323	0.243
not sure	uncertainty	0.037	0.218	0.181
I assume	uncertainty	0.02	0.068	0.048
not entirely	uncertainty	0.006	0.048	0.042
could be	uncertainty	0.103	0.144	0.041
possibly	uncertainty	0.0	0.008	0.008
reconsider	second_guess	0.0	0.0	0.0
on second thought	second_guess	0.0	0.0	0.0
presumably	uncertainty	0.009	0.006	-0.002
I guess	uncertainty	0.12	0.049	-0.07
wait/actually no	second_guess	0.413	0.341	-0.072
actually	second_guess	0.391	0.312	-0.079
but wait	second_guess	0.687	0.528	-0.159
hold on	second_guess	0.741	0.476	-0.265
let me recheck	second_guess	0.852	0.474	-0.378
double-check	second_guess	0.864	0.474	-0.39
hmm	second_guess	2.566	2.078	-0.488

Table 7: Doubt density (markers per 1000 reasoning words) split by final outcome, per category (2g = second.guess, un = uncertainty, tot = both). For a group, density = total markers in that group / total reasoning words in that group x 1000. *_correct = over correct-answer traces; *_wrong = over wrong-answer traces; *_wrong_over_correct = _wrong / _correct.

config	2g_correct	2g_wrong	2g_wrong_over_correct	un_correct	un_wrong	un_wrong_over_correct	tot_correct	tot_wrong	tot_wrong_over_correct
clean	11.859	16.213	1.367	5.674	12.973	2.286	17.534	29.186	1.665
typo25_real10	12.006	14.085	1.173	8.361	18.69	2.235	20.367	32.775	1.609
typo25_real40	11.867	11.896	1.002	7.809	15.34	1.964	19.675	27.236	1.384
typo25_real70	11.948	13.848	1.159	8.095	18.085	2.234	20.043	31.933	1.593
typo50_real10	11.46	12.724	1.11	9.389	16.722	1.781	20.849	29.446	1.412
typo50_real40	11.234	11.935	1.062	10.751	18.932	1.761	21.985	30.867	1.404
typo50_real70	11.909	13.275	1.115	10.988	17.502	1.593	22.896	30.777	1.344
typo75_real10	11.075	14.785	1.335	10.716	21.556	2.012	21.79	36.341	1.668
typo75_real40	10.383	13.467	1.297	11.038	23.135	2.096	21.422	36.602	1.709
typo75_real70	11.454	11.661	1.018	15.155	24.941	1.646	26.609	36.601	1.376

Repair Behaviour

Table 8: How each corrupted word was handled, per config (categories defined below the table).

config	n_corrupt_words	silent_fix_n	flagged_n	misread_n	not_used_n	silent_fix_frac	flagged_frac	misread_frac	not_used_frac
typo25_real10	3668	2623	370	105	570	0.7151	0.1009	0.0286	0.1554
typo25_real40	3657	2504	461	143	549	0.6847	0.1261	0.0391	0.1501
typo25_real70	3669	2379	578	160	552	0.6484	0.1575	0.0436	0.1504
typo50_real10	7181	4850	987	283	1061	0.6754	0.1374	0.0394	0.1478
typo50_real40	7155	4672	1063	320	1100	0.653	0.1486	0.0447	0.1537
typo50_real70	7169	4249	1310	414	1196	0.5927	0.1827	0.0577	0.1668
typo75_real10	9898	5769	2120	550	1459	0.5828	0.2142	0.0556	0.1474
typo75_real40	9758	5592	1989	613	1564	0.5731	0.2038	0.0628	0.1603
typo75_real70	9737	4932	2240	814	1751	0.5065	0.2301	0.0836	0.1798

A *corrupted word* is a word that received a typo — it differs between the clean and typo question (e.g. *sum* → *sun*). We recover these by diffing the two questions, then place each into one bucket by which form appears in the reasoning:

- **silent_fix** — only the original word (*sum*) appears: a quiet recovery.
- **flagged** — both forms appear (*sum* and *sun*): noticed and fixed.
- **misread** — only the corrupted form (*sun*) appears: read as the wrong word.
- **not_used** — neither form appears in the reasoning.

Columns *_n are counts; *_frac are the share of n_corrupt_words.

Table 9: Word handling pooled by real-word ratio: the non-word vs real-word contrast.

real_ratio	n_corrupt_words	silent_fix_frac	flagged_frac	misread_frac	not_used_frac
10	20747	0.6383	0.1676	0.0452	0.1489
40	20570	0.6207	0.1708	0.0523	0.1562
70	20575	0.5618	0.2006	0.0675	0.1701

Table 10: Misread rate split by final correctness. A typo’d word is ‘misread’ when the reasoning uses only its corrupted form (e.g. sun instead of sum). Within each outcome group, misread rate = (typo’d words that were misread) / (all typo’d words examined), pooled over the group. misread_correct = over problems answered correctly; misread_wrong = over problems answered wrong; wrong_over_correct = misread_wrong / misread_correct.

config	misread_correct	misread_wrong	wrong_over_correct
typo25_real10	0.0261	0.0417	1.598
typo25_real40	0.0344	0.0602	1.75
typo25_real70	0.0414	0.0524	1.267
typo50_real10	0.0369	0.0469	1.273
typo50_real40	0.0412	0.0525	1.273
typo50_real70	0.0526	0.0671	1.275
typo75_real10	0.0492	0.0675	1.372
typo75_real40	0.0575	0.0705	1.225
typo75_real70	0.0764	0.0915	1.197

Real-word Effect

This section isolates the effect of typo *kind* (real-word vs non-word), holding the question, corrupted positions, and rate fixed — something the vs-clean accuracy section cannot do. Three views:

- a **controlled paired** comparison of low-real vs high-real variants on the *same* questions, with a McNemar test (paired table);
- a **per-typo logistic regression** comparing the marginal harm of one real-word vs one non-word typo (logit table);
- whether wrong answers **flag the corruption** more or less as the real-word share rises (silent-failure table).

Read the signs, deltas and p -values in the tables below for this dataset’s result.

Table 11: Controlled paired comparison of two TYPO variants against EACH OTHER (baseline is the low-real variant, NOT clean – so this is not the vs-clean accuracy table). Within a fixed rate the same questions are corrupted in the same positions, only the typo KIND differs. Restricted to questions answered in both variants. acc_low_real / acc_high_real = accuracy on the lower / higher real-word variant; delta_acc = acc_high_real - acc_low_real; r2w = questions correct in the low-real variant but wrong in the high-real one; w2r = the reverse; mcnemar_p = McNemar test on that paired change. Negative delta with r2w $\hat{>}$ w2r means real-word typos are more harmful than the non-word typos they replaced.

rate	compare	n_both	acc_low_real	acc_high_real	delta_acc	r2w	w2r	mcnemar_p
25	real10_to_real70	498	0.8373	0.8032	-0.0341	43	26	0.05408269
25	real10_to_real40	496	0.8407	0.8306	-0.0101	38	33	0.63499179
25	real40_to_real70	496	0.8286	0.8044	-0.0242	45	33	0.21294628
50	real10_to_real70	498	0.7631	0.6807	-0.0823	86	45	0.00047439
50	real10_to_real40	498	0.7631	0.7169	-0.0462	70	47	0.04196161
50	real40_to_real70	498	0.7149	0.6787	-0.0361	64	46	0.105042
75	real10_to_real70	484	0.6674	0.5702	-0.0971	102	55	0.00024141
75	real10_to_real40	482	0.6701	0.6369	-0.0332	75	59	0.19504349
75	real40_to_real70	482	0.6432	0.5768	-0.0664	80	48	0.00614319

Table 12: Logistic regression $\text{correct} \sim \text{num_real} + \text{num_nonword}$ over all answered typo traces ($\text{num_real} / \text{num_nonword}$ = count of real-word / non-word typos in the problem). coef_* = change in log-odds of a correct answer per one added typo of that kind (more negative = more harmful); $\text{odds_*} = \exp(\text{coef})$; intercept is the model constant. n = number of traces fit.

n	coef_num_real	coef_num_nonword	odds_real	odds_nonword	intercept
4465	-0.0743	-0.03665	0.9284	0.964	1.79061

Table 13: Among WRONG answers only, does the model flag the corruption, by real ratio? $\text{explicit_notice_frac}$ = fraction of wrong-answer traces whose reasoning uses strong typo-flagging language (typo / misspelled / doesn’t make sense / ...); $\text{silent_frac} = 1 - \text{explicit_notice_frac}$. The proposal predicts real-word failures are quieter ($\text{explicit_notice_frac}$ should fall as real rises).

real_ratio	n_wrong	explicit_notice_frac	silent_frac
10	362	0.6298	0.3702
40	404	0.6312	0.3688
70	471	0.6178	0.3822

Lexical Grid (overview)

Table 14: All four marker families (second_guess , uncertainty , typo_noticing , repair_words) across configs.

config	second_guess	uncertainty	typo_noticing	repair_words
clean	12.641	6.985	1.939	1.42
typo25_real10	12.603	11.327	3.828	2.673
typo25_real40	11.875	10.033	3.861	2.492
typo25_real70	12.6	11.524	4.308	2.991
typo50_real10	11.926	12.092	4.772	3.25
typo50_real40	11.576	14.744	5.072	3.418
typo50_real70	12.574	14.158	4.923	3.397
typo75_real10	12.991	16.314	5.224	3.437
typo75_real40	12.226	18.267	5.608	4.095
typo75_real70	11.577	20.985	5.891	4.127